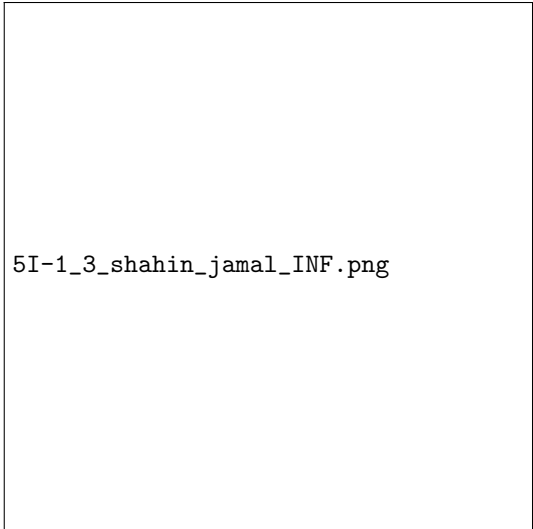


Auteur: Shahin Jamal
Studierichting: Informatica
Oefeningenreeks 5I nr. 1.3

$$\begin{aligned}y &= x^2 - \frac{1}{8} \ln x \Rightarrow y' = 2x - \frac{1}{8x} \Rightarrow (y')^2 = 4x^2 - \frac{1}{2} + \frac{1}{64x^2} \\ \Rightarrow 1 + (y')^2 &= 1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2} = \frac{256x^4 + 32x^2 + 1}{64x^2} \\ \int_2^3 \sqrt{\frac{256x^4 + 32x^2 + 1}{64x^2}} dx &= \int_2^3 \sqrt{\frac{(16x^2 + 1)^2}{64x^2}} dx = \int_2^3 \frac{16x^2 + 1}{8x} dx \\ &= \int_2^3 \frac{16}{8} x + \frac{1}{8x} dx = \left[x^2 + \frac{1}{8} \ln|x| \right]_2^3 = 9 + \frac{1}{8} \ln|3| - \left(4 + \frac{1}{8} \ln|2| \right) \\ &= 5 + \frac{1}{8} \ln \frac{3}{2}\end{aligned}$$



5I-1_3_shahin_jamal_INF.png

References